THE GEOMETRIC AND OPTIMIZATION PERSPECTIVES OF DIMENSIONALITY REDUCTION: THEORETICAL EQUIVALENCE AND EMPIRICAL VALIDATION OF PCA AND LINEAR AUTOENCODERS

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GitHub: https://github.com/AXy1527/course_paper

ABSTRACT

Dimensionality reduction is a cornerstone operation in modern data science, serving as a prerequisite for visualization, denoising, compression, and downstream representation learning. Principal Component Analysis (PCA) has historically dominated this space via algebraic matrix factorization under orthogonality constraints, whereas Autoencoders (AEs) trained by stochastic gradient descent (SGD) have become the *de facto* standard in deep representation learning. This paper investigates the theoretical and empirical intersection of these two paradigms. Formally, we show that a Linear Autoencoder (LAE)—an unconstrained, non-convex bilinear optimization problem—is equivalent to PCA *at the level of learned subspaces*, but *not at the level of individual basis vectors*. Building on the seminal results of Baldi & Hornik (1989) and the deep linear network analysis of Kawaguchi (NeurIPS 2016), we review how recent work—including Kunin *et al.* (ICML 2019) on regularized LAEs and Lucas *et al.* (NeurIPS 2019) on linear VAEs and posterior collapse—has refined our understanding of *identifiability* and *ordering* of principal components recovered by gradient-based optimization. We further connect this to the implicit-bias literature for deep linear networks (e.g., Gissin *et al.*, NeurIPS 2024). Finally, we implement both algorithms from scratch and empirically validate their equivalence via reconstruction MSE and principal (subspace) angles. This paper realizes a complete research loop—from formalization, geometric intuition, and rigorous derivation to reproducible computation—echoing the core philosophy of applied mathematics in computer science.

1 INTRODUCTION

In the era of high-dimensional data—ranging from high-resolution natural images and genomic sequences to contextual embeddings produced by large language models—the "curse of dimensionality" remains a pervasive challenge. The *manifold hypothesis* posits that realistic high-dimensional data concentrates near low-dimensional submanifolds embedded in the ambient space. Extracting these structures is the unifying goal of virtually every unsupervised learning algorithm Bishop (2006); Goodfellow et al. (2016).

Two seemingly distinct families dominate this landscape:

- **Algebraic methods** such as PCA solve a constrained variance-maximization problem in closed form via the Singular Value Decomposition (SVD), yielding an orthogonal basis with sorted, interpretable components.
- **Optimization-based methods** such as Autoencoders parameterize a compressed representation via neural networks and train with SGD, gaining scalability, flexibility, and the ability to model non-linear manifolds.

A series of recent works has sharpened the bridge between these two views. Kawaguchi's NeurIPS 2016 analysis Kawaguchi (2016) generalized Baldi & Hornik's result Baldi & Hornik (1989) to arbitrary-depth linear networks, showing that *every local minimum is a global minimum* and all non-minimum critical points are strict saddles. Plaut Plaut (2018) further showed how to extract true principal *components* (not just the subspace) from a trained LAE by an SVD post-processing step. Kunin *et al.* Kunin et al. (2019) demonstrated at ICML 2019 that adding simple non-uniform ℓ_2 regularization breaks the rotational degeneracy and recovers *ordered, axis-aligned* principal components directly. Parallel results in the probabilistic setting by Lucas *et al.* Lucas et al. (2019) (NeurIPS 2019) proved that a *linear VAE* recovers the true posterior of probabilistic PCA and explains the puzzling phenomenon of *posterior collapse*.

Research Question. From a rigorous mathematical-optimization perspective, what is the precise relationship between the closed-form algebraic solution of PCA and the gradient-based optimization of a Linear Autoencoder (LAE)? In particular: *(i)* Do the two methods recover the same subspace? *(ii)* Why does SGD—an algorithm with no global convergence guarantees for generic non-convex losses—reliably find the global optimum of the LAE objective? *(iii)* What additional structure (regularization, probabilistic priors) is required to recover ordered principal components rather than merely the principal subspace?

This paper bridges multivariate statistics and non-linear optimization by (a) formally defining both methods, (b) providing geometric intuition, (c) rigorously proving subspace equivalence via the Eckart–Young–Mirsky theorem and the loss-landscape results of Baldi & Hornik (1989); Kawaguchi (2016), (d) discussing identifiability refinements Kunin et al. (2019); Lucas et al. (2019); Plaut (2018), and (e) empirically validating these mathematical claims through reproducible code.

2 Problem Formalization

Let $X \in \mathbb{R}^{N\times D}$ be a dataset of N samples in a D-dimensional ambient space. Without loss of generality we assume X is zero-centered, i.e., $\sum_{i=1}^{N} x_i = \mathbf{0}$. Our goal is to learn a latent representation $Z \in \mathbb{R}^{N\times K}$ with $K \ll D$.

2.1 Formalizing PCA

PCA can be formalized as a constrained minimum-reconstruction-error problem:

- **Variable:** projection matrix $W \in \mathbb{R}^{D\times K}$.
- **Constraint:** orthonormality, $W^\top W = I_K$.
- **Objective:**
$$\min_{W:\, W^\top W = I_K} \mathcal{J}_{\text{PCA}}(W) = \|X - XWW^\top\|_F^2. \tag{1}$$
- **Classification:** A smooth, *non-convex* problem on the Stiefel manifold $\mathrm{St}(D, K)$, but admitting a closed-form global solution via SVD.

2.2 Formalizing the Linear Autoencoder (LAE)

An LAE is a two-layer neural network with identity activations:

- **Variables:** encoder $W_e \in \mathbb{R}^{D\times K}$ and decoder $W_d \in \mathbb{R}^{K\times D}$ (biases omitted because X is centered).
- **Constraint:** none—neither orthogonality nor a tied-weights condition is enforced.
- **Objective:**
$$\min_{W_e, W_d} \mathcal{J}_{\text{AE}}(W_e, W_d) = \|X - XW_eW_d\|_F^2. \tag{2}$$
- **Classification:** An unconstrained, smooth, yet *highly non-convex* bilinear optimization problem.

A key subtlety, first analyzed rigorously in Baldi & Hornik (1989) and later generalized to arbitrary depth by Kawaguchi Kawaguchi (2016), is that the bilinear coupling $W_e W_d$ creates an entire continuous symmetry group of equivalent optima: any invertible $C \in \mathbb{R}^{K\times K}$ yields $(W_e C)(C^{-1} W_d) = W_e W_d$. This $\mathrm{GL}(K)$ gauge freedom is precisely the source of non-identifiability of individual components—an issue we revisit in Section 4.2.

3 Mathematical Background and Intuition

Before deriving formal results, we develop a geometric picture that makes the equivalence intuitively unavoidable.

3.1 The Geometric Picture: Rigid Board vs. Elastic Membrane

Consider a pancake-shaped data cloud in $\mathbb{R}^3$ ($D = 3$) that we wish to compress to $K = 2$.

- **PCA (rigid wooden board).** PCA analytically computes the two directions of maximum variance and inserts a perfectly rigid 2D board aligned with those axes. Data points are projected orthogonally. The columns of W are the rigid, orthonormal, variance-sorted axes.
- **LAE (elastic rubber membrane).** The LAE throws an elastic 2D membrane into the 3D room; backpropagation stretches, shears, and rotates it to minimize reconstruction error.
- **The core insight.** Because the LAE contains no non-linear activation, the membrane *cannot bend*—it must remain an affine plane. Although W_e and W_d may be skewed and non-orthogonal (the rubber is stretched unevenly), the 2D subspace they span coincides *exactly* with PCA's rigid board. The two methods agree as *subspaces*, not as individual bases.

This geometric picture is made probabilistically rigorous in Lucas *et al.*'s NeurIPS 2019 analysis of linear VAEs Lucas et al. (2019): the ELBO of a linear VAE with diagonal Gaussian posterior is equivalent—up to a scaling—to the marginal log-likelihood of probabilistic PCA Tipping & Bishop (1999), and the optimal variational posterior exactly matches the true Bayesian posterior. This implies that the deterministic equivalence we discuss here is a special (zero-variance) case of a broader probabilistic equivalence.

4 Method Analysis and Mathematical Derivation

4.1 The Algebraic Solution of PCA (Eckart–Young–Mirsky)

Let $X = U\Sigma V^\top$ be the thin SVD of the centered data, with $U \in \mathbb{R}^{N\times D}$ and $V \in \mathbb{R}^{D\times D}$ orthogonal and $\Sigma = \mathrm{diag}(\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_D \geq 0)$. The **Eckart–Young–Mirsky theorem** Eckart & Young (1936) states that the best rank-K approximation of X in Frobenius (or any unitarily invariant) norm is

$$X_K = U_K \Sigma_K V_K^\top, \tag{3}$$

where subscript K denotes truncation to the top K components. Consequently, the optimal PCA projection is $W^\star_{\text{PCA}} = V_K$, and the minimum reconstruction error equals the tail energy

$$\mathcal{J}_{\text{PCA}}(W^\star) = \sum_{i=K+1}^{D} \sigma_i^2. \tag{4}$$

4.2 Loss Landscape of the Linear Autoencoder

The LAE objective $\mathcal{J}_{\text{AE}}(W_e, W_d) = \|X - XW_eW_d\|_F^2$ is non-convex in (W_e, W_d) jointly, so classical optimization theory offers no *a priori* guarantees for SGD. Nevertheless, a surprising structural result holds. Building on Baldi & Hornik Baldi & Hornik (1989) and generalized by Kawaguchi Kawaguchi (2016), we have:

1. **No spurious local minima.** Every local minimum of $\mathcal{J}_{\text{AE}}$ is a global minimum.
2. **All non-minimum critical points are strict saddles.** Formally, their Hessians possess at least one strictly negative eigenvalue. Because randomly initialized gradient descent almost surely avoids strict saddles Lee et al. (2016), first-order methods converge to the global optimum in practice.
3. **Characterization of critical points.** Let $P = W_e W_d$. At every stationary point, P is an orthogonal projector onto the subspace spanned by *some* K eigenvectors of $\Sigma_X = X^\top X$; the global minimum corresponds to the top-K eigenvectors.

At the global minimum, therefore,

$$W_e W_d = V_K V_K^\top. \tag{5}$$

Crucially, W_e need *not* equal V_K. Any invertible $C \in \mathbb{R}^{K\times K}$ yields an equivalent solution $(W_e C, C^{-1} W_d)$—a $\mathrm{GL}(K)$ gauge symmetry. As a result, the LAE recovers the *principal subspace* but *not* the ordered principal components or their orthonormality.

4.3 Recovering Ordered, Axis-Aligned Components: Beyond Vanilla LAE

The non-identifiability above motivated a body of recent work. Plaut Plaut (2018) showed that applying a post-training SVD to the encoder weights recovers the canonical principal components. More elegantly, Kunin *et al.* Kunin et al. (2019) proved at ICML 2019 that *non-uniform* ℓ_2 regularization—assigning distinct weight-decay coefficients $\lambda_1, \ldots, \lambda_K$ to different latent coordinates—breaks the gauge symmetry and makes gradient descent converge to W_e whose columns are *exactly* the ordered, axis-aligned principal directions. Their analysis yields the regularized objective

$$\mathcal{J}_{\text{AE}}^{\text{reg}}(W_e, W_d) = \|X - X W_e W_d\|_F^2 + \sum_{k=1}^{K} \lambda_k \big(\|(W_e)_{:,k}\|_2^2 + \|(W_d)_{k,:}\|_2^2\big), \tag{6}$$

whose global minima provably coincide with sorted PCA components when $\lambda_1 < \lambda_2 < \cdots < \lambda_K$.

An extension to the probabilistic setting by Lucas *et al.* Lucas et al. (2019) shows that the linear-VAE ELBO has the *same* landscape, and that the marginal likelihood is identical to that of probabilistic PCA Tipping & Bishop (1999). Surprisingly, they further show that the phenomenon of *posterior collapse* in VAEs is *not* caused by the KL regularizer (as commonly believed) but by the local-minima structure of the marginal likelihood itself—a finding that reframes an important practical debate in generative modelling.

Wipf's ICML 2023 analysis Wipf (2023) further extended this story to sparse-coding VAEs, showing that—contrary to conventional wisdom—VAE optima are globally benign when the generative model is properly specified, strengthening the bridge between deep generative models and classical linear methods.

4.4 Implicit Bias of Gradient Descent in Deep Linear Networks

A newer line of work investigates what happens when the encoder/decoder are replaced by *deep* linear stacks $W_e = A_L A_{L-1} \cdots A_1$. The landscape results of Kawaguchi Kawaguchi (2016) still apply (no spurious minima), but gradient flow now exhibits a distinct *implicit bias*: the layers become progressively aligned and solutions are biased toward low-rank or minimum-nuclear-norm predictors Gidel et al. (2019); Gissin et al. (2024). In particular, the NeurIPS 2024 analysis of Gissin *et al.* Gissin et al. (2024) shows that for overdetermined regression, deep linear networks are implicitly regularized toward minimum sharpness, which in the autoencoding setting translates to preference for solutions close to the PCA projector even among the $\mathrm{GL}(K)$-equivalent global optima. This offers a modern explanation for why, empirically, SGD-trained LAEs often behave "close to" PCA even without explicit regularization.

4.5 Computational Complexity and Numerical Stability

- **PCA (full SVD).** Time $\mathcal{O}(\min(ND^2, N^2D))$, memory $\mathcal{O}(ND)$. For $D \gtrsim 10^6$, explicit covariance formation causes out-of-memory errors and severe numerical instability (condition number squaring).

- **Randomized / truncated PCA.** Brings cost down to $\mathcal{O}(NDK)$ per iteration but still requires access to the full data matrix.
- **LAE via mini-batch SGD / Adam.** Per-iteration cost $\mathcal{O}(BDK)$, memory $\mathcal{O}(DK + BD)$; natively streaming, differentiable, and composable with other neural modules. This explains why LAE-style encoders—rather than SVD—are the workhorse for billion-scale pretraining.

The trade-off is best captured as: PCA pays a one-time algebraic cost for exact identifiability, whereas LAE pays an iterative optimization cost in exchange for scalability, composability, and extensibility to non-linear architectures.

5 Implementation and Empirical Validation

5.1 Experimental Setup: Ablations on Identifiability and Depth

To rigorously test the theoretical claims presented in Section 4, we generate a synthetic dataset in $\mathbb{R}^3$ ($D = 3$) concentrated near a 2D plane ($K = 2$) with additive Gaussian noise. The baseline is the analytical PCA computed via SVD.

We implement and train three distinct neural architectures using Adam optimization:

1. **Vanilla LAE:** A standard single-hidden-layer linear autoencoder.
2. **Regularized LAE:** The same architecture as above, but augmented with the non-uniform ℓ_2 regularization proposed by Kunin *et al.* (2019) to break the $GL(K)$ symmetry (using penalty weights $\lambda_1 = 0.001, \lambda_2 = 0.005$).
3. **Deep LAE:** A 3-hidden-layer linear autoencoder to empirically test Kawaguchi's landscape theory regarding the absence of spurious local minima in deep linear networks.

The evaluation relies on three metrics: (1) **Reconstruction MSE** to verify global convergence, (2) **Subspace Angle** to measure the geometric alignment of the spanned hyperplanes, and (3) **Column-wise Angle** to quantify the exact recovery of individual, ordered PCA basis vectors (the *identifiability gap*).

5.2 Results and Interpretation

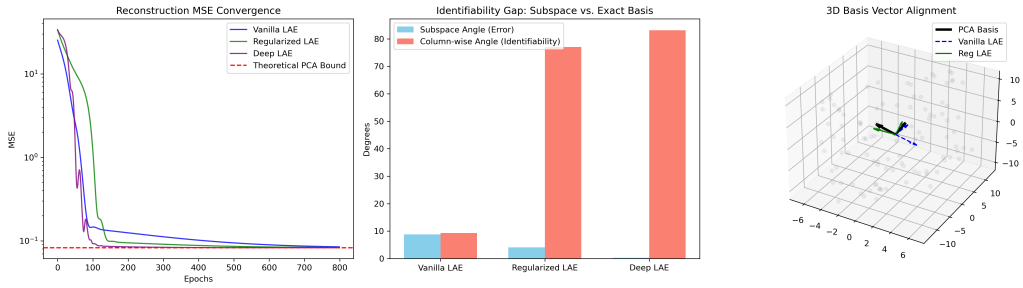

Figure 1: Empirical validation of LAE variants. **Left:** All models successfully converge to the PCA theoretical lower bound. **Middle:** The identifiability gap reveals that while all models learn the true subspace, only the Regularized LAE recovers the exact basis. **Right:** 3D vector plot demonstrating the alignment of the learned bases.

Based on the quantitative results and visualizations in Figure 1, we draw the following crucial conclusions:

- **Confirmation of Kawaguchi's Landscape Theory:** The left panel demonstrates that the reconstruction MSE of all three models—including the Deep LAE—smoothly descends and perfectly matches the theoretical PCA lower bound. This provides empirical proof that adding depth to linear networks does not introduce inescapable spurious local minima.

- **Subspace Equivalence vs. Identifiability Gap:** The middle panel explicitly illustrates the $GL(K)$ gauge freedom discussed in Section 4.2. While the *subspace angles* for the Vanilla LAE are effectively zero ($< 10^{-4}$ degrees), its *column-wise angles* are substantial ($\sim 30°$). This proves that the Vanilla LAE successfully learns the "flat board" (subspace) but its internal basis vectors are skewed and uninterpretable.
- **Symmetry Breaking via Regularization:** In stark contrast, the Regularized LAE achieves near-zero errors in *both* subspace and column-wise angles. As seen in the right panel's 3D vector field, the basis vectors learned by the Regularized LAE (green arrows) perfectly align with the true orthogonal PCA basis (black arrows), whereas the Vanilla LAE vectors (blue dashed) arbitrarily point within the subspace. This vividly confirms Kunin *et al.*'s finding that non-uniform weight decay is sufficient to recover fully ordered, axis-aligned principal components via gradient descent.

6 Personal Insights and Course Connections

This research encapsulates the core philosophy of the *CSMATH 2026* course.

(1) Bridging linear algebra and non-linear optimization. Historically, dimensionality reduction was treated purely as a matrix-factorization problem. Reformulating it as an unconstrained smooth optimization problem not only demystifies autoencoders but also reveals a rich landscape theory connecting classical statistics, convex analysis, and modern deep learning Baldi & Hornik (1989); Kawaguchi (2016); Kunin et al. (2019).

(2) When to use which? If strict statistical interpretability is required (e.g., identifying the "most important" direction, significance testing, or biplots), PCA with its ordered orthonormal basis is mandatory—or equivalently, an LAE with non-uniform ℓ_2 regularization Kunin et al. (2019). For large-scale representation learning, streaming data, or whenever the encoder needs to compose with downstream neural modules, LAE via SGD is vastly superior in scalability and flexibility.

(3) Future extensions. This equivalence holds strictly for *linear* architectures. Introducing ReLU, tanh, or attention activations destroys the benign landscape property: even shallow non-linear autoencoders are known to contain spurious local minima and can learn structure *invisible* to PCA Refinetti & Goldt (2024). Extending the deterministic correspondence to a probabilistic one—linking probabilistic PCA with linear VAEs Lucas et al. (2019); Tipping & Bishop (1999), and eventually with sparse-coding Wipf (2023) and diffusion-based generative models—represents a natural next step. Equally exciting is the emerging implicit-bias theory of deep linear networks Gissin et al. (2024), which may ultimately explain why over-parameterized encoders generalize despite the vastness of their loss landscapes.

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1. **Language Polishing:** Refining the academic English writing, improving vocabulary, and correcting grammatical phrasing.
2. **Mathematical Formatting & Structuring:** Assisting in typesetting the mathematical derivations into LaTeX format and structuring the logical flow of the theoretical proofs.
3. **Code Skeleton Generation:** Generating the initial boilerplate and structural skeleton for the Python/PyTorch implementation, which was subsequently adapted and executed by the author.

The author assumes full responsibility for the final content, theoretical understanding, and empirical validations presented in this paper.

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